

Path Smoothing and Turning-Maneuver Optimization for Differential-Drive Mobile Robots

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Abstract—Global planners in ROS 2/Nav2 commonly produce piecewise-linear paths with abrupt heading changes, resulting in high curvature peaks and reduced motion continuity for differential-drive mobile robots. This paper proposes PSTMO, a path-smoothing and turning-maneuver optimization method. In contrast to greedy line-of-sight shortcuts, which may remove anchors that encode important path features and the intended approach direction, standalone PSTMO does not execute LOS. The planner path is first conditioned under bounded-deviation and swept-footprint constraints, after which only local corner regions are modified. At each corner, the method derives at most two geometrically grounded trim distances d : a preferred value based on the adjacent edge lengths and a compatibility value that reserves at most half of the usable shared segment after a margin. For each d , the dimensionless ratio $\alpha=q/d$ is searched on a coarse grid from 0.1 to 0.5, refined between the neighbors of the winning sample, and recovered using a half-step grid only when all coarse samples fail. A quintic G^2 Bézier transition is admitted only if it satisfies geometric validity, non-reversing inner-wheel motion, kinematic limits, timing constraints, and true swept-footprint collision checks. A time gate and dynamic programming then select among transitions and in-place rotations while rejecting states whose trims overlap on a shared segment.

The geometric evaluation contains 175 records from 7 Gazebo environments and 5 global planners: NavFn A*, NavFn Dijkstra, Theta*, Smac 2D, and Smac Hybrid. Five alternatives are evaluated: Raw, Simple, Savitzky–Golay, Constrained, and PSTMO. Within each case, all alternatives receive the same Raw path. PSTMO returned a valid output in all 35/35 calls, passed the final safety checks, and produced no footprint-collision samples. Raw, Savitzky–Golay, and Constrained also succeeded in 35/35 cases, whereas Simple succeeded in 34/35. Across the 34 cases completed by all five alternatives, PSTMO achieved a mean path length of 10.1766 m, maximum curvature of 2.4047 m^{-1} , and squared-curvature integral of 3.3121 m^{-1} . Relative to Simple, which attained the lowest mean baseline value for all three corresponding metrics, these quantities decreased by 0.64%, 75.29%, and 81.44%, respectively. Relative to Constrained, maximum curvature and squared-curvature integral decreased by 80.62% and 92.10%.

Travel time was measured in a separate closed-loop experiment over the same 7 environments using Theta*. Each alternative was executed once per environment in a fresh simulation, from the same start pose to the same goal. Time was measured from controller path acceptance until the robot reached the goal and physically settled according to Gazebo ground truth. All 35/35 executions reached the goal without collision-monitor intervention. Mean travel times for Raw, Simple, Savitzky–Golay, Constrained, and PSTMO were 60.465 s, 55.635 s, 59.005 s, 60.545 s, and 55.813 s, respectively. PSTMO reduced mean travel time by 7.69% relative to Raw, 5.41% relative to Savitzky–Golay, and 7.82% relative to Constrained, while being 0.32% slower than Simple. These results show that PSTMO produces short, low-curvature paths while retaining near-best travel time. Repeated trials with additional planners and physical hardware remain future work.

Keywords—PSTMO; differential-drive mobile robot; path smoothing; turning-maneuver optimization; G^2 Bézier transition; coarse-to-fine search; swept footprint; ROS 2/Nav2.